



Md Mahbub Ali

Master's of Science Student,
Energy Engineering

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Bangladeshi

Social Network

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Language

English

Beginner Danish – Sprog Center
(course begins 27 Mar 2025)

Skills

PSCAD

MATLAB Simulink

PowerFactory

MS Office, Google Editors

About Me

I am an international student at Aalborg University, pursuing a Master's in Energy Engineering, specializing in Electrical Power Systems and High Voltage Engineering. Currently, I am working on my thesis on VSG-based grid-forming converter control systems, focusing on power system large-signal transient stability analysis.

As I near the completion of my studies, I am actively seeking job opportunities where I can apply my skills, gain hands-on experience, and contribute to the industry. My primary goal is to become a grid engineer, and I am deeply passionate about renewable energy and power systems. However, I am open to exploring any role related to electrical power systems.

Project Experience

- **Current Thesis Work:** Transient Stability Analysis of Multi-VSG Low-Inertia Power Grids

Studying virtual synchronous generator based grid-forming control for low-inertia grids.

Simulating a 9-bus power system single-line diagram with VSG-GFM control for large-signal transient analysis in PSCAD.

- **Third Semester Project:** Black-Start Capability Analysis via Grid Forming Control

Studied droop-based grid-forming control system application.

Simulated an 80MVA single-line diagram in PSCAD using open source data from Eurowind Energy Company to supply a 20MW/15MVAR load 30s post-blackout via a grid-forming converter.

- **Second Semester Project:** Machine Learning Based Power Flow Analysis

Studied the use of Graph Neural Networks to optimize Newton-Raphson-based power flow analysis, achieving faster computation compared to traditional methods.

Validated machine learning analysis data using PowerFactory.

- **First Semester Project:** Fractional Frequency Transmission System for Off-shore Wind Power.

Designing and simulating a three-phase cyclo converter for bidirectional frequency conversion.

- **Embedded Electronics Internship (Jan,2022 - Jun,2022):** Aqualink Bangladesh Limited

Developed protective, digital, and analog control circuits for advanced electronic systems.

Technical experience in installing and operating I/O controllers on industrial power panels to control exhaust centrifugal fans via VFD.

- **Bachelor Thesis:** Conducted a theoretical energy accumulation study on a two-level dual-axis bi-facial module-based PV system.

Education

Current(4th Semester) **MSc. in Energy Engineering**

Aalborg University

- Studied power system stability analysis, advanced power electronics, High voltage system, electrical Machines and Control Systems.

Jan,2016 - Dec,2020 **BSc. in Electrical and Electronic Engineering**

BRAC University

- Studied power plant engineering, energy conversion, and power electronics.